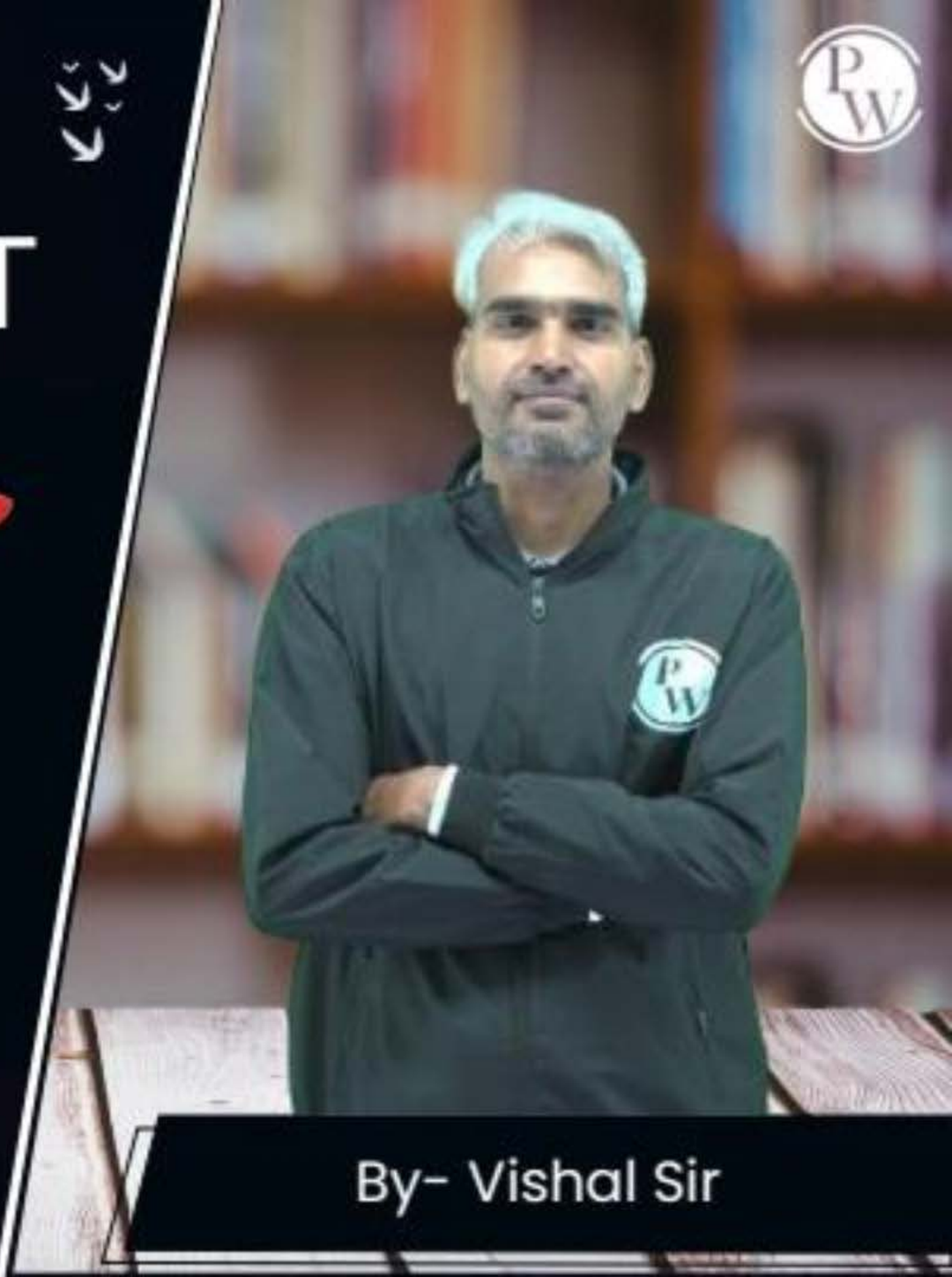
Computer Science & IT



Database Management System

Query Languages

Lecture No. 12



By- Vishal Sir

Recap of Previous Lecture



- Topic: Practice questions
- Topic: Introduction to relational calculus
- Topic: Tuple relational calculus

Slide

Topics to be Covered



- Topic — Practice questions
- Topic — Domain relational calculus (DRC)

Slide

H.W.

#Q. Consider a database that has the relation schemas

- EMP(EmpId, EmpName, DeptId) and
- DEPT(DeptName, DeptId).

Note that the DeptId can be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

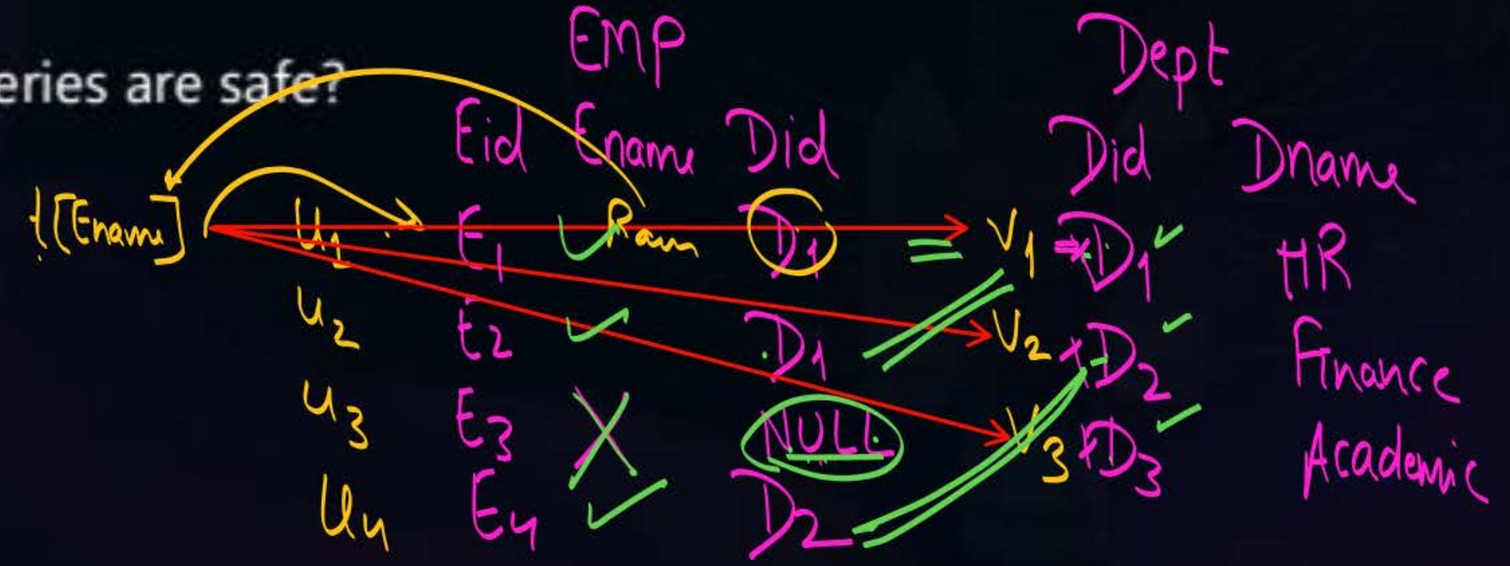
I. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \forall v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

II. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

III. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] = DeptId]))\}$

Which of the above queries are safe?

A. I and II only
B. I and III only
C. II and III only
D. I, II and III



H.W:

#Q. [MCQ: 2009: 2M]

Consider the following relational schema:

Suppliers(sid:integer, sname:string, city:string, street:string)

Parts(pid:integer, pname:string, color:string)

Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database:

```
SELECT S.sname
        FROM Suppliers S
        WHERE S.sid NOT IN
                (SELECT C.sid
                        FROM Catalog C
                        WHERE C.pid NOT IN
                                (SELECT P.pid
                                        FROM Parts P
                                        WHERE P.color<> \'blue\'))
```

i.e. Sids of Suppliers who supplied only non-blue parts or no parts

o/p will also include the Sids of Suppliers who did not supply any part

Sids of Suppliers who did not supply any blue color parts

Sids of Suppliers who have supplied some blue color parts

It will select the Pids of all non-blue color parts

o/p = S_2-Sname, S_4-Sname

o/p = S_1, S_3

Supplier

Sid
S_1
S_2
S_3
S_4

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_3
S_3	P_2

Parts

Pid	Color
P_2	Blue
P_1	Red
P_3	Green

o/p = P_1, P_3

Question Continues in Next Slide

#Q. Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

A. ✗ Find the names of all suppliers who have supplied a non-blue part.

B. ✗ Find the names of all suppliers who have not supplied a non-blue part.

C. ✗ Find the names of all suppliers who have supplied only blue parts.

D. ✗ Find the names of all suppliers who have not supplied only blue parts.

non

Suppliers who supplied some other Color parts along with at least one blue Color part

Domain Relational Calculus



"Column"



Topic : Syntax of DRC

{ t | P(t) }

Columns (Attribute)

- Each query is an expression of the form

  {<x1,x2,....,xn> | P(x1,x2,.....,xn) }

  - Where x1,x2,.....,xn represent output domain variables.
  - P(x1,x2,.....,xn) is the predicate formula.

Consider the following relation

Emp (Eid, Ename, Salary)

Retriere all records of the Emp table

TRC $\{ t \mid t \in Emp \}$

DRC: $\{ \langle E, N, S \rangle \mid \langle E, N, S \rangle \in Emp \}$

We need all three Column in o/p

'E' is the name assigned to 1st Column of Emp

'N' is the name assigned to 2nd Column of Emp

S is the name assigned to 3rd Column of Emp

Consider the following relation

Emp (Eid, Ename, Salary)

Retriere Ename of all Employee

TRC $\{ t.Ename \mid t \in Emp \}$

DRC: $\{ \langle N \rangle \mid \langle E, N, S \rangle \in Emp \}$

We need Column w.r.t. Ename {i.e. 2nd Column}

'E' is the name assigned to 1st Column of Emp

'N' is the name assigned to 2nd Column of Emp

S is the name assigned to 3rd Column of Emp

Consider the following relation

Emp (Eid, Ename, Salary)

Retriere Eid of all Employee

TRC $\{ t.Eid \mid t \in Emp \}$

DRC: $\{ \langle \bar{E} \rangle \mid \langle \bar{E}, \bar{N}, \bar{S} \rangle \in Emp \}$

'E' is the name assigned to 1st Column of Emp

'N' is the name assigned to 2nd Column of Emp

S is the name assigned to 3rd Column of Emp

#Q. Consider the following relational schemas
Employee(EmpId, EmpName, salary)

Select distinct EmpName from Employee table where salary > 2000

TRC: $\{ t.EmpName \mid t \in Employee \wedge t.Salary > 2000 \}$

DRC: $\{ \langle N \rangle \mid \langle E, N, S \rangle \in Employee \wedge \langle S \rangle > 2000 \}$

#Q. Consider the following relational schemas
Student(Rollno, name, marks)

Select distinct name and marks of the student from Student table where Rollno = 10

TRC : $\{t.name, t.marks \mid t \in Student \wedge t.Rollno = 10\}$

DRC: $\{ \langle N, M \rangle \mid \langle R, N, M \rangle \in Student \wedge \langle R \rangle = 10 \}$

(or)

$\{ \langle N, M \rangle \mid \langle 10, N, M \rangle \in Student \}$

#Q. Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?

"Find the distinct names of all students who score more than 90% in the course numbered 107"

$$R.A = \Pi_{S.sname}\left(\sigma_{R.CourseNo=107 \wedge R.percent>90}\left(Student(S) \times Registration(R)\right)\right)$$

$$R.RollNo = S.RollNo.$$

SQL

```
Select distinct S.Sname
From Student S, Registration R
Where ( R.CourseNo = 107
        AND
        R.Percent > 90
        ANP
        R.RollNo = S.rollno)
```

TRC :-

$$\{ t.Sname \mid t \in Student \wedge \exists R \in Registration \; (R.CourseNO = 107 \wedge R.Percent > 90 \wedge R.rollno = t.rollno) \}$$

$$\{ t \mid \exists S \in Student \; (\exists R \in Registration \; (R.rollno = S.rollno \wedge R.CourseNO = 107 \wedge R.Percent > 90) \; (\text{or}) \; t.Sname = S.Sname) \}$$

I. SELECT DISTINCT S.sname FROM Students as S, Registration as R WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90

II. $\prod_{sname}(\sigma_{courseno=107 \wedge percent > 90}(Registration \bowtie Students))$

III. $\{ T \mid \exists S \in Students, \exists R \in Registration (S.rollno=R.rollno \wedge R.courseno=107 \wedge R.percent > 90 \wedge T.sname=S.sname)\}$

IV. $\{ < S_N > \mid \exists S_R \exists R_P (< S_R, S_N > \in Students \wedge < S_R, 107, R_P > \in Registration \wedge R_P > 90)\}$

A. I, II, III and IV

B. I, II and III only

C. I, II and IV only

D. II, III and IV only



2 mins Summary

Topic Practice questions

Topic Domain relational calculus (DRC)

Slide

THANK - YOU